

```
import java.util.Scanner;
```

```
public class Q1{  
    public static void main(String[] args){  
        Scanner sc =new Scanner(System.in);  
        System.out.println("Enter how many number You want to print:");  
        int N = sc.nextInt();  
        for(int i=1;i<=N;i++){  
            System.out.println(i);  
        }  
    }  
}
```

```
import java.util.Scanner;
```

```
public class Q2{
```

```
    public static void main(String[] args){
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter a number:");
```

```
        int N =sc.nextInt();
```

```
        for(int i=1;i<=N;i++){
```

```
            int result=3*i;
```

```
            System.out.println(result);
```

```
        }
```

```
    }
```

```
}
```

```
import java.util.Scanner;
```

```
public class Q3{
```

```
    public static void main(String[] args){
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter a number:");
```

```
        int N=sc.nextInt();
```

```
        long factorial=1;
```

```
        for (int i=1;i<=N;i++){
```

```
            factorial=factorial*i;
```

```
        }
```

```
        System.out.println("Factorial of "+N+"="+factorial);
```

```
    }
```

```
}
```

```
import java.util.Scanner;
```

```
public class Q4{
```

```
    public static void main(String args[]){
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a number:");
```

```
        int N=sc.nextInt();
```

```
        for(int i=1;i<=N;i++){
```

```
            if(i%2==0){
```

```
                System.out.println(i);
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

```
import java.util.Scanner;
```

```
public class Q5{  
    public static void main(String[] args){  
        Scanner sc=new Scanner(System.in);  
        System.out.println("Enter a number:");  
        int N=sc.nextInt();  
        int sum=0;  
        for(int i=1;i<=N;i++){  
            if(i%2!=0){  
                sum=sum+i;  
            }  
        }  
        System.out.println("Sum:"+sum);  
    }  
}
```

```
import java.util.Scanner;
```

```
public class Q6 {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int[] numbers = new int[5];  
  
        System.out.println("Enter 5 integers:");  
  
        for (int i = 0; i < 5; i++) {  
            numbers[i] = sc.nextInt();  
        }  
  
        System.out.println("Array elements are:");  
  
        for (int num : numbers) {  
            System.out.println(num);  
        }  
  
        sc.close();  
    }  
}
```

```
import java.util.Scanner;
```

```
public class Q7 {
```

```
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);
```

```
        int[] numbers = new int[5];
```

```
        System.out.println("Enter 5 integers:");
```

```
        for (int i = 0; i < 5; i++) {  
            numbers[i] = sc.nextInt();  
        }
```

```
        int sum = 0;
```

```
        for (int num : numbers) {  
            sum = sum + num;  
        }
```

```
        System.out.println("Sum of all elements = " + sum);
```

```
        sc.close();
```

```
    }
```

```
}
```

```
import java.util.Scanner;
```

```
public class Q8 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        String[] names = new String[4];
```

```
        System.out.println("Enter 4 names:");
```

```
        for (int i = 0; i < 4; i++) {
```

```
            names[i] = sc.nextLine();
```

```
        }
```

```
        System.out.println("Names are:");
```

```
        for (String name : names) {
```

```
            System.out.println(name);
```

```
        }
```

```
        sc.close();
```

```
    }
```

```
}
```



```
import java.util.Scanner;  
import java.util.Arrays;
```

```
public class Q9 {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int[] numbers = new int[5];  
  
        System.out.println("Enter 5 integers:");  
  
        for (int i = 0; i < 5; i++) {  
            numbers[i] = sc.nextInt();  
        }  
  
        Arrays.sort(numbers);  
  
        System.out.println("Largest element = " + numbers[4]);  
  
        sc.close();  
    }  
}
```

```
import java.util.Scanner;

public class Q10 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];

        System.out.println("Enter 5 integers:");

        for (int i = 0; i < 5; i++) {
            numbers[i] = sc.nextInt();
        }

        int sum = 0;

        for (int num : numbers) {
            sum = sum + num;
        }

        double average = (double) sum / numbers.length;

        System.out.println("Average = " + average);

        sc.close();
    }
}
```

```
import java.util.Scanner;
```

```
public class Q11 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int[] numbers = new int[6];
```

```
        System.out.println("Enter 6 integers:");
```

```
        for (int i = 0; i < 6; i++) {
```

```
            numbers[i] = sc.nextInt();
```

```
        }
```

```
        int positive = 0;
```

```
        int negative = 0;
```

```
        for (int num : numbers) {
```

```
            if (num > 0) {
```

```
                positive++;
```

```
            } else if (num < 0) {
```

```
                negative++;
```

```
            }
```

```
        }
```

```
        System.out.println("Number of positive numbers = " + positive);
```

```
        System.out.println("Number of negative numbers = " + negative);
```

```
        sc.close();
```

```
    }
```

```
}
```

```
import java.util.Scanner;
import java.util.Arrays;

public class Q12 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];

        System.out.println("Enter 5 integers:");

        for (int i = 0; i < 5; i++) {
            numbers[i] = sc.nextInt();
        }

        Arrays.sort(numbers);

        System.out.println("Sorted array in ascending order:");

        for (int num : numbers) {
            System.out.println(num);
        }

        sc.close();
    }
}
```

```
import java.util.Scanner;
import java.util.Arrays;

public class Q13 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        Integer[] numbers = new Integer[5];

        System.out.println("Enter 5 integers:");

        for (int i = 0; i < 5; i++) {
            numbers[i] = sc.nextInt();
        }

        System.out.print("Enter number to search: ");
        int search = sc.nextInt();

        if (Arrays.asList(numbers).contains(search)) {
            System.out.println("Found");
        } else {
            System.out.println("Not Found");
        }

        sc.close();
    }
}
```

```
import java.util.Scanner;
import java.util.Arrays;

public class Q14 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];

        System.out.println("Enter 5 integers:");

        for (int i = 0; i < 5; i++) {
            numbers[i] = sc.nextInt();
        }

        // binarySearch() requires the array to be sorted
        Arrays.sort(numbers);

        System.out.print("Enter number to search: ");
        int search = sc.nextInt();

        int index = Arrays.binarySearch(numbers, search);

        if (index >= 0) {
            System.out.println("Index = " + index);
        } else {
            System.out.println("Not Found");
        }

        sc.close();
    }
}
```

```
public class Q16 {  
    public static void main(String[] args) {  
  
        for (int i = 1; i <= 5; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print(j);  
  
                if (j < i) {  
                    System.out.print("*");  
                }  
            }  
            System.out.println();  
        }  
    }  
}
```



```
public class Q17 {  
    public static void main(String[] args) {  
  
        for (int i = 1; i <= 5; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print(2 * j - 1);  
  
                if (j < i) {  
                    System.out.print(" ");  
                }  
            }  
            System.out.println();  
        }  
    }  
}
```



```
public class Q18 {  
    public static void main(String[] args) {  
        for (int i = 1; i <= 5; i++) {  
            for (int j = 1; j <= 5; j++) {  
                System.out.print(i);  
            }  
            System.out.println();  
        }  
    }  
}
```

```
public class Q19 {  
    public static void main(String[] args) {  
        for (int i = 1; i <= 5; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print(i);  
            }  
            System.out.println();  
        }  
    }  
}
```

```
public class Q20 {  
    public static void main(String[] args) {  
  
        for (int i = 1; i <= 5; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print(j);  
            }  
            System.out.println();  
        }  
    }  
}
```

```
public class Q21 {  
    public static void main(String[] args) {  
  
        int num = 1;  
  
        for (int i = 1; i <= 5; i++) {  
            for (int j = 1; j <= i; j++) {  
                System.out.print(num + " ");  
                num++;  
            }  
            System.out.println();  
        }  
    }  
}
```

```
public class Q22 {  
    public static void main(String[] args) {  
  
        for (int i = 1; i <= 6; i++) {  
            for (int j = 1; j <= 6; j++) {  
  
                if (i == 1 || i == 6 || j == 1 || j == 6) {  
                    System.out.print("*");  
                } else {  
                    System.out.print(" ");  
                }  
            }  
            System.out.println();  
        }  
    }  
}
```

```
public class Q23 {  
    public static void main(String[] args) {  
  
        for (int i = 1; i <= 6; i++) {  
  
            for (int j = 1; j <= 6 - i; j++) {  
                System.out.print(" ");  
            }  
  
            for (int j = 1; j <= 2 * i - 1; j++) {  
                System.out.print("*");  
            }  
  
            System.out.println();  
        }  
  
        for (int i = 5; i >= 1; i--) {  
  
            for (int j = 1; j <= 6 - i; j++) {  
                System.out.print(" ");  
            }  
  
            for (int j = 1; j <= 2 * i - 1; j++) {  
                System.out.print("*");  
            }  
  
            System.out.println();  
        }  
    }  
}
```

```
import java.util.Scanner;
```

```
public class Q24 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a string: ");
```

```
        String str = sc.nextLine();
```

```
        String reverse = "";
```

```
        for (int i = str.length() - 1; i >= 0; i--) {
```

```
            reverse = reverse + str.charAt(i);
```

```
        }
```

```
        System.out.println("Reversed string: " + reverse);
```

```
        sc.close();
```

```
    }
```

```
}
```

```
import java.util.Scanner;
```

```
public class Q25 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a string: ");
```

```
        String str = sc.nextLine();
```

```
        int count = 0;
```

```
        for (int i = 0; i < str.length(); i++) {
```

```
            char ch = Character.toLowerCase(str.charAt(i));
```

```
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
```

```
                count++;
```

```
            }
```

```
        }
```

```
        System.out.println("The number of vowels in '" + str + "' is: " + count);
```

```
        sc.close();
```

```
    }
```

```
}
```



```
import java.util.Scanner;
```

```
public class Q26 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a string: ");
```

```
        String str = sc.nextLine();
```

```
        String reverse = "";
```

```
        for (int i = str.length() - 1; i >= 0; i--) {
```

```
            reverse = reverse + str.charAt(i);
```

```
        }
```

```
        if (str.equals(reverse)) {
```

```
            System.out.println("The string '" + str + "' is a palindrome.");
```

```
        } else {
```

```
            System.out.println("The string '" + str + "' is not a palindrome.");
```

```
        }
```

```
        sc.close();
```

```
    }
```

```
}
```

```
public class Q27 {  
    public static void main(String[] args) {  
  
        String str1 = "hello";  
        String str2 = "hello";  
  
        System.out.println("Both variables point to the same object: " + (str1 == str2));  
    }  
}
```

```
public class Q28 {  
    public static void main(String[] args) {  
  
        String str1 = new String("hello");  
        String str2 = new String("hello");  
  
        System.out.println("Using ==: " + (str1 == str2));  
        System.out.println("Using equals(): " + str1.equals(str2));  
    }  
}
```

```
public class Q29 {  
    public static void main(String[] args) {  
  
        String str1 = "hello";  
        String str2 = "world";  
  
        String str3 = str1 + str2;  
  
        System.out.println("Is str3 pointing to the same object as str1: " + (str3 == str1));  
    }  
}
```

```
public class Q30 {  
    public static void main(String[] args) {  
  
        String str1 = new String("hello");  
        String str2 = str1.intern();  
        String str3 = "hello";  
  
        System.out.println("Is str2 and str3 pointing to the same object? " + (str2 == str3));  
    }  
}
```

```
public class Q31 {  
    public static void main(String[] args) {  
  
        String str1 = "java";  
        String str2 = "java";  
        String str3 = "java";  
  
        System.out.println("All strings point to the same object: " +  
                           (str1 == str2 && str2 == str3));  
    }  
}
```